ETAS RTA-CAR 12.7.0

RTA-CAR Release Notes

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ETAS RTA-CAR 12.7.0 - RTA-CAR Release Notes R01 EN – 05.2025

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1 Introduction

1.1 Document References

- ETAS Safety Advice, ETAS GmbH
- RTA-CAR User Guide, 12.7.0, ETAS GmbH
- RTA-CAR Release Notes, 12.7.0, ETAS GmbH
- ISOLAR-A Getting Started Guide, 12.7.0, ETAS GmbH
- ISOLAR-A Reference Manual, 12.7.0, ETAS GmbH
- ISOLAR-A Release Notes, 12.7.0, ETAS GmbH
- ISOLAR-B Getting Started Guide, 12.7.0, ETAS GmbH
- ISOLAR-B Reference Manual, 12.7.0, ETAS GmbH
- ISOLAR-AB Release Notes, 12.7.0, ETAS GmbH
- RTA-BSW Reference Manual, 12.7.0, ETAS GmbH
- RTA-BSW Release Notes, 12.7.0, ETAS GmbH
- RTA-RTE Getting Started Guide, 12.7.2, ETAS GmbH
- RTA-RTE User Guide, 12.7.2, ETAS GmbH
- RTA-RTE Reference Manual, 12.7.2, ETAS GmbH
- RTA-RTE Release Notes, 12.7.2, ETAS GmbH
- RTA-OS Getting Started Guide, 12.6.0, ETAS GmbH
- RTA-OS User Guide, 12.6.0, ETAS GmbH
- RTA-OS Reference Manual, 12.6.0, ETAS GmbH
- RTA-OS Release Notes and specific Port Guide, 12.6.0, ETAS GmbH
- MemMapGen Getting Started Guide, 12.7.0, ETAS GmbH
- MemMapGen Reference Manual, 12.7.0, ETAS GmbH
- RTA-A2L Getting Started Guide, 12.7.0, ETAS GmbH
- RTA-A2L Reference Manual, 12.7.0, ETAS GmbH

1.2 User Documentation

All product documentation in PDF format can be found in the Delivery Package.

2 Product Definition

2.1 Functions at a Glance

RTA-CAR provides a common installer for delivering all ETAS products in the RTA Classic Autosar Solution, and it offers an integrated environment for configuring and generating ECU source code for the Classic AUTOSAR stack.

RTA-CAR focuses on:

- The configuration of the Application Layer.
- The configuration and generation of the Run Time Environment (RTE).
- The configuration and generation of the Basic Software Layer (BSW).
- The configuration and generation of the Operating System (OS).

The RTA-CAR solution includes the following ETAS tools:

- ISOLAR-A 12.7.0: AUTOSAR Authoring Tool that assists users in designing application software to AUTOSAR standards.
- ISOLAR-B 12.7.0: BSW configuration tool.
- RTA-BSW 12.7.0: BSW automatic configuration and generation.
- RTA-RTE 12.7.2: Run Time Environment generation.
- RTA-OS 12.6.0: Operating System generation.
- RTA-A2L 12.7.0: A2L generator for use with measurement and calibration tools.
- MemMapGen 12.7.0: Memory map generator.
- LiMa 1.8.12.61: ETAS License Manager.

Please refer to the tool-specific documentation for details on their features.

2.2 General Description

2.2.1 System Prerequisites

The following is the recommended execution environment:

2.2.1.1 Windows

Recommended Hardware	- 2.5 GHz Quad-Core,x86 PC or equivalent - 16 GB RAM - Graphics with a resolution of at least 1920 x 1080 - 10 GB Free disk space - Network card (NIC)

Operating System	- Windows® 11 64-bit - Windows® 10 64-bit - Windows Server® 2022 64-bit - Windows Server® 2019 64-bit

2.2.1.2 Linux

Recommended Hardware	- 2.5 GHz Quad-Core, x86 PC or equivalent - 16 GB RAM - Graphics with a resolution of at least 1920 x 1080 - 10 GB Free disk space - Network card (NIC)
Operating System	- Linux® Ubuntu 22.04 64-bit - Linux® Ubuntu 20.04 64-bit

2.2.2 Software Prerequisites

RTA-CAR does not have any software prerequisites.

2.2.3 Target Prerequisites

RTA-CAR BSW requires a target where aligned memory accesses to 8, 16 and 32 bit variables is always atomic. This requirement does not apply to auxiliary cores like digital signal processors (DSPs) and coprocessors.

2.3 Delivery

The software delivery package includes the RTA-CAR software, documentation, tools, utilities, and further information.

The software is packaged as an executable installer (Windows) or as a .deb package with an accompanying installation script (Linux).

Please see the RTA-CAR User Guide for installation instructions.

Documentation is provided in the Portable Document Format (PDF).

2.3.1 OSS and third-party software usage

RTA-CAR uses open source components, refer to the OSS attribution documents for details. RTA-CAR does not use any other third-party software.

2.4 Installation

See the RTA-CAR User Guide for detailed information on the installation of RTA-CAR.

2.5 Licensing

RTA-CAR requires you to have the following valid licenses for the individual components of RTA-CAR.

ISOLAR-A	- ISOLAR_A V12
ISOLAR-B	- ISOLAR_B V12
RTA-BSW	- LD_RTA_BSW V12 (see note 1) - LD_RTA_BSW_LIN V12 (see note 2) - LD_RTA_BSW_ETH V12 (see notes 2 and 3) - LD_RTA_BSW_ETH_SEC V12 (see notes 2) - LD_RTA_BSW_FRAY V12 (see note 2) - LD_RTA_BSW_J1939 V12 (see note 2) - LD_RTA_BSW_SEC V12 (see note 2) - LD_RTA_BSW_SWCL V12 (see note 2) - LD_RTA_BSW_RIPSPLUGINS V12 (see note 3)
RTA-RTE	- RTA-RTE V12 - RTA-RIPS V12 (enables the RIPS functionality, see note 3) - RTA-A2L V7
RTA-OS	- ES_RTA-IOC V5 (enables the IOC Optimization feature) - LD_RTA-OS_VRTA_SC34 V5 (enables the OS tools and VRTA-win) - LD_RTA-OS_SRC V5 (enables the output of debug source files) - A target specific license is also required
RTA-A2L	- RTA-A2L V12

Notes:

1. The only mandatory license for RTA-BSW is LD_RTA_BSW, which covers the licensing for all the core RTA-BSW modules that are standard across most AUTOSAR projects.
2. The other RTA-BSW licenses are optional, and their requirement is based on the type of communications used within the project (e.g. Ethernet, Lin, FlexRay, J1939) and the inclusion of optional features such as Security and SWCL.
3. The LD_RTA_BSW_ETH license, which is required if your project uses

Ethernet communications, also includes the LD_RTA_BSW_RIPSPLUGINS and RTA-RIPS licenses.

2.6 Compatibility

RTA-CAR ensures compatibility between the following ETAS products and product versions.

Product	Version
ISOLAR-A	12.7.0
ISOLAR-B	12.7.0
RTA-BSW	12.7.0
RTA-RTE	12.7.2
RTA-OS	12.6.0
RTA-A2L	12.7.0
MemMapGen	12.7.0
LiMa	1.8.12.61

3 Release Changes

3.1 Changes and Defects Fixed

This section lists the major changes that provide significant improvements to RTA-CAR workflows or add support for new AUTOSAR modules.

For information on other changes present in the tool versions included in this release, please refer to the release notes for the specific tool.

3.1.1 Key Improvements to RTA-CAR 12.7.0

The following changes have been made:

3.1.1.1 This version of RTA-CAR introduces the following new major changes:

- Improvements to support the simplified integration of BSW Exclusive Areas. BSW modules now provide information on the exclusive areas they require, allowing the automatic mapping of a lock implementation by the RTE generator.
- New FeeFs3 module. This variant of the Fee module provides a data layout that is optimized for overwritable memories like Phase-Change or Resistive Random Access Memories (PCM/RRAM).
- RTA-CAR is updated to use Java 21 and Eclipse version 2024-09 (4.33).
- The RTA-CAR Windows installer now includes a bundled virtual starter kit to help users get up and running with a worked example project.
- Support for the generation of SBoM (Software Bill of Materials) information alongside the BSW code generated by RTA-CAR, containing information about the modules that were included in the code generation process.

3.1.1.2 Diagnostic and Calibration Enhancements

- Extensions to DCM service 0x29 to support Renault use cases.
- Support configurations where DemMaxNumberFreezeFrame is 0, as per AUTOSAR [ECUC_Dem_00605].
- Support DcmDspDataByteSize parameter as defined in AUTOSAR 4.3 and newer.
- Support for UDS Service 0x86 (Response on Event) sub-functions 0x07 (enable event type) and 0x09 (disable event type).
- Support of DCM_OEM_BOOT_RESPAPP and DCM_SYS_BOOT_RESPAPP values for DcmDspSessionForBoot parameter
- Support use of FIM in a multi-core or multi-partition environment with new master-satellite architecture.
- UDS service $34 support for memorySize and memoryAddress as per ISO-14229-1:2020-02.

3.1.1.3 Communication Enhancements

- CanIf support for mixed mode operation (use of standard and extended CAN IDs on the same physical channel, as per [SWS_CANIF_00281]).
- Introduce templates for bus-specific type definitions, to allow adaptation in line with project requirements.
- Support for multi-ASIL use cases.
- Support for the Nagle Algorithm in TCP/IP, according to IETF RFC 896.
- Support SOME/IP Access Control Lists (ACL) according to AUTOSAR R24-11.
- Support graceful close of TCP sockets during ECU reset.

3.1.1.4 Tooling Enhancements

- ConfGen new automatically creates a default Det configuration where it is identified as a dependency.
- Enhancements to version migration feature.
- Improved standard workflow for Ethernet configuration
- Improvements to BSW and BswM editors
- Improvements to DBC importer to support import of frames as SecuredIP-dus. Note that some manual configuration is necessary as the DBC file does not contain all the necessary information needed to allow fully automatic configuration.
- Improvements to ISOLAR model validations
- Improvements to LDF importer to support signal groups, E2E and FrameTriggerings
- New editor for the configuration of exclusive areas.
- Unified command line interface for ASW and BSW use cases, across Windows and Linux platforms.

3.1.1.5 Crypto and Security Enhancements

- Extensions to CycurHSM3 crypto driver.
- Improvements to key management operations when supported by an underlying CycurHSM2 implementation.
- Multithreading enhancements in CryptoAuCSC.
- Provide separate APIs for HKDF Extract and Expand operations.
- Support for TLS_PSK_WITH_AES_128_GCM_SHA256 and TLS_PSK_WITH-_CHACHA20_POLY1305_SHA256 cipher suites.
- Support input and output buffering to automatically handle buffer alignment.
- Support the use of 3072-bit RSA keys.

- Support X.509v3 certificate handling using an underlying CycurHSM3 implementation.
- Various certificate handling improvements.

3.1.1.6 Run Time Environment Enhancements

- Generation of measurement buffers for unconnected PRPorts is now supported.
- Initial support for CAT2 ISRs, as per requirements [SWS_Rte_07515] and [SWS_Rte_04563] in AUTOSAR R23-11.
- Support for DataReceivedEvents that are not mapped to an OsTask.
- Support for the typeEmitter attribute as per AUTOSAR requirements [SWS_Rte_06712] and [SWS_Rte_04632].
- Support mapping of reentrant server RunnableEntitys to OsTasks.
- Support NetworkRepresentation in conjunction with data transformation (non-AUTOSAR implementation using adapter SWC).

3.1.1.7 Other Component Improvements

- New memory allocation keyword VAR_INIT_LOCK_SEMA to support isolation of locks and semaphores.
- Support for E2E profile P07m.

The following components have been updated:

- ISOLAR-A/B
- MemMapGen
- RTA-A2L
- RTA-BSW
- RTA-RTE

For more details, please see the relevant component Release Notes.

3.2 Known Issues

3.2.1 Windows Specific

3.2.1.1 Executing ISOLAR-AB

In order to launch ISOLAR-AB, the command line executable (ISOLAR-AB.exe) must be run from the ISOLAR-AB installation directory.

3.2.1.2 Re-using an ISOLAR-AB workspace

The default ISOLAR-AB workspace will not be the same if ISOLAR-AB.exe is executed from different locations. For example, if you first execute

ISOLAR-AB.exe from the standard installed folder, and you then later execute the same instance of the executable from a command line from another location, the original workspace in the installed folder will not be used, and instead a brand new workspace will be created in the folder from where ISOLAR-AB.exe was executed.

3.2.2 Linux Specific

3.2.2.1 Executing ISOLAR-AB

In order to launch ISOLAR-AB, the command line executable (ISOLAR-AB) must be run from the ISOLAR-AB installation directory.

3.2.2.2 Re-using an ISOLAR-AB workspace

The default ISOLAR-AB workspace will not be the same if ISOLAR-AB is executed from different locations. For example, if you first execute ISOLAR-AB from the standard installed folder, and you then later execute the same instance of the executable from a command line from another location, the original workspace in the installed folder will not be used, and instead a brand new workspace will be created in the folder from where ISOLAR-AB was executed.

3.2.2.3 RTA-OS VRTA Port Availability on Linux

The RTA-OS VRTA port is currently unavailable on Linux, therefore this port cannot selected when creating a project.

Current Workaround - The Linux version includes an VRTA-UX port, which can be used to generate projects instead of the RTA-OS VRTA port.

3.2.2.4 GUI Issues

When using RTA-CAR in the graphical user interface mode some windows may exhibit strange graphical artefacts. These are only present on a limited number of pages and are not relevant to the use of RTA-CAR.

3.2.2.5 Permissions Discrepancies

The primary RTA-CAR installation directory (`/usr/share/ETAS/RTA-CAR_12.7.0`) does not have the expected permissions (read-only) but is instead writable by all users matching behaviour on Windows.

3.2.3 All Platforms

3.2.3.1 ECUExtract settings

Instead of obeying ECUExtract UI preference settings, Confgen generation considers settings from a text file (EcuExtract_CLI_Input.txt), which needs to be available within the project.

3.2.3.2 Other Known Issues

Please refer to the Release Notes for the specific products for any other known issues.

3.3 Hotfix Information

Please refer to the Release Notes for the specific products.

3.4 Release History

3.4.1 Version 12.6.0

3.4.1.1 New Features

- Initial support for “Live Validation” at configuration time. This allows real-time validation of parameters as configuration is performed, providing faster feedback and allowing errors to be corrected more quickly. Not all parameters are supported in this release, support will be extended in future versions.
- RTA-CAR now supports the migration of a RTA-CAR project created in an earlier version (starting from V12.3.0) to the current RTA-CAR version (12.6.0). This migrates project configurations along with project-specific settings. Users are provided with a summary of changes in the project and a guide mentioning specific configurations that needs minimal user intervention.
- Support for CanXL as specified in AUTOSAR R22-11.

3.4.1.2 Diagnostic and Calibration Enhancements

- Automatic configuration of Dem NvBlock sizes.
- Extensions to freeze frame storage for Heavy Duty vehicles (OnPendingAndUpdateOnConfirmed).
- Post-Build selectable support for DCM parameters DcmDslProtocolRowUsed and DcmDsdServiceUsed.
- Support for calibration of the packed bit-coded parameters DemDtcValue, DemDtcValue3Byte, DemDebounceCounterPassedThreshold, DemDebounceCounterFailedThreshold, DemRbDebounceCounterSuspiciousThreshold and DemRbDebounceCounterFdcThresholdStorageValue.
- Support for Scheduler type #2 in UDS service 0x2A.
- Support for the AUTOSAR parameter DcmDspPidUsed.
- Support service $19 sub-functions $1A and $56 in the standard UDS context.
- Support use of Diagnostic stack (DEM and DET) in multicore/multi-partition configurations.
- DoIP v4 support for TP Status payload (0x9001) according to ISO_13400-2_AMD_1_2023-07.

3.4.1.3 Communication and System Stack Enhancements

- Support use of WdgM in multicore/multi-partition configurations.

- Buffer pooling support for multicast messages.
- Support wildcards for supplier ID and function ID according to ISO 17987-3:2016.
- The RTA-BSW PduR module has been extended so that PDU groups that have the parameter 'PduRRbEnableSeqTransmit' set to true will transmit the PDUs in those groups in the same order as they are received. This allows PDU order to be maintained in the gateway use case.
- WdgIf support for external watchdog drivers.
- Support for handling ComSignals and LdComIPdus across different partitions.
- Optimizations and improvements to gateway use cases.

3.4.1.4 Tooling Enhancements

- Automatic generation of schedule configuration based on execution ordering constraints.
- Improved Ethernet configuration workflow.
- Improvements to ISOLAR-A/B validations.
- Improvements to the ISOLAR-A/B command line interface.
- Performance improvements with large projects.
- Usability and performance improvements in BSW Editor.
- Workflow improvements in DEXT editor.
- ConfGen support for TLS configuration.

3.4.1.5 Crypto and Security Enhancements

- Updated key storage handling to utilize new CycurSoC APIs.
- IdsM offers measurement points to give insights into event reception, qualification and transmission statistics as well as buffer fill level and fill distribution.
- Improvements to memory allocation in line with modern AUTOSAR versions.
- KeyM now supports the use of OSCP and CRL in parallel for X.509 certificates.
- New module CryptoAuHSM3 provides a CryptoDriver compatible with CycurHSM3.
- SecOC support for verification status callouts.
- SHE key handling is extended to support up to 100 symmetric keys.
- Support for flexible usage of callback functions and CSM jobs.
- Support for IPSec with Authentication Header (AH) & Internet Key Exchange (IKE).

- TLS support for TLS_ECDHE_ECDSA_WITH_AES_128_CCM_8 and Record Size Limit Extension.

3.4.1.6 Other Component Improvements

- Optimizations to resource locking to improve runtime performance.
- Support for Fee page size alignment via paramdef.
- Support use of NvM stack in multicore/multi-partition configurations.
- New RTA-CAR Security Case Summary document to support conformance with ISO 21434.
- Support automatic size configuration for NvM block lengths.
- Support for RIPS invocation handlers to be mapped to non-periodic tasks.
- Support for unconnected R-ports (where RPortPrototype.mayBeUnconnected is set) in ASW components.

3.4.2 Version 12.5.0

- Memstack now supports the Flash EEPROM Emulation 2.0 (FeeFs2) module that emulates the possibilities of a serial EEPROM inside a flash memory.
- Generic Importers supports the import of a subset of Service Needs to configure the diagnostic stack.
- UDS Services 0x14 (Clear Diagnostic Information), 0x2A (Read Data By Periodic Identifier) are updated in accordance with ISO 14229-1:2020-02.
- Diagnostics is updated to support the following features from AUTOSAR 23-11, S3 Server timeout is configurable; API Dem_SetEventConfirmationThresholdCounter is supported.
- OBD services as per updated specification SAE J1979DA_202305 are now supported.
- StbM is extended to allow TSyn modules to report security events to the IdsM.
- NvM_SetRamBlockStatus can now be disabled via configuration.
- The RTA-RTE now supports Signal fan-IN for Sender-Receiver with Data Transformation for Com.
- Configuration of CompuMethod is now under SystemSignal instead of ISignal.
- CSSafety RIPS plugin now supports a counter-based timeout handling instead of OsAlarmCallBack, This improves runtime performance of the generated code.
- PostBuildSelectable is now supported in BSWM module for container BswMRule and the sub containers BswMRuleTrueActionList / BswMRuleFalseActionList.
- E2E Profile P08, P04m are now supported.

- E2E protection of client/server communication (Data Transformation approach using Rte_Rips_CSXfrm RIPS plugin) is now supported.
- Various Useability improvements to ISOLAR-B OS & BSWM Editors.
- J1939 Service DM26 is now supported.
- Handshake Functionality of TLS 1.3 is supported.
- The Time Service module is now integrated and supported.

3.4.3 Version 12.4.0

- ARXML files serialized by ISOLAR-A/B now use UTF-8 character encoding.
- Calibration with XCP is now supported in the J1939 Stack.
- CERT C L1 compliance statements are now provided for RTA-BSW, RTA-OS and RTA-RTE.
- Enhancements to the Ecu Partition Editor to improve the process of mapping software components to partitions.
- Extensions to authentication service $29 in the DCM module.
- For inter-ECU communication, auto-generated CompuMethods are now attached to the SystemSignal and not to the mapped ISignal.
- IdsM Security Event reporting is now supported in the following modules: BswM, CanIf, Csm, Dcm, Dem, EthIf, KeyM, PduR and SecOC.
- Improved support for split ParamDef files in ConfGen.
- Improvements to the Ethernet configuration workflow.
- ISOLAR-A/B now supports validation of CPFlex configurations.
- MemMapGen now uses the Vendor ID of a BSW/MCAL component when generating the corresponding header files.
- MemMapGen now validates pragma configurations.
- RTA-A2L can now merge customer A2L record layouts with different Alignment sizes.
- RTA-A2L now supports generation of elements of datatype float without data constraints.
- RTA-CAR now supports ISO 15118-20 cryptographic primitives for V2G applications.
- RTA-CAR now supports the communication of DoIP Lite diagnostic message payloads.
- The Com stack now supports Bus Mirroring to replicate the traffic and state of internal buses to an external bus.
- The Com stack now supports the simultaneous use of multiple CAN drivers.
- The DCM module now supports the use of multiple diagnostic protocols in parallel in a default session.
- The DEM module now provides a configuration parameter to tune the

number of IUMPR ratios that are processed on each invocation of the main function.

- The Diagnostic stack now supports SAE J1979-3 for Zero Emission vehicles.
- The DoIP module now allows the `DoIPTcpConnectionSecurityRequired` parameter to be overridden at runtime, for cases where TLS certificate generation needs to be invoked via a diagnostic routine.
- TLS now includes support for the TLS_ECDHE_ECDSA_WITH-_AES_128_GCM_SHA256, TLS_ECDHE_ECDSA_WITH-_AES_256_GCM_SHA384, TLS_ECDHE_ECDSA_WITH-_AES_128_CBC_SHA256, TLS_ECDHE_ECDSA_WITH-_AES_256_CBC_SHA384, and TLS_ECDHE_ECDSA_WITH_NULL_SHA cipher suites.
- Usability enhancements to BswM editor.
- Usability enhancements to the OS editor.
- Various enhancements to the Rte_Rips_CSSafety module.
- Various improvements to the DBC importer.

3.4.4 Version 12.3.2

- The PduR module has been extended so that PDU groups that have the parameter `PduRRbEnableSeqTransmit` set to true will transmit the PDUs in those groups in the same order as they are received. This allows the PDU order to be maintained in the gateway use case.
- The EthTrcv module now supports AUTOSAR CONC_632 wakeup and sleep over dataline.
- For Tx PDUs with 'dynamic' length, `PduR_SecOCTransmit` will now be called with no padding bytes. Behavior for Tx PDUs with 'static' length is unchanged.
- For service-based traffic, SecOC will now use the FVID calculated at runtime from the KSN received in the payload instead of the configured FVID.
- Dcm module parameters `DcmDspPeriodicTransmissionSlowRate`, `DcmDspPeriodicTransmissionMediumRate`, and `DcmDspPeriodicTransmissionFastRate` have been updated so that their max values match the AUTOSAR 20-11 specification.

3.4.5 Version 12.3.1

- Multicast network traffic can now be routed to the XCoreCdd component to support gateway use cases.
- Support for client/server timeouts in Rte_Rips_CSXfrm without the use of OsAlarmCallback.
- RTA-RTE now generates an OsScheduleTable for each core when the Ecuc parameter RteGeneration.RteScheduleTablePerCore is set to true.

- Enhancements to queueing of multicast data in the PduR component to support gateway use cases.
- Additional documentation has been added for vendor-specific parameters used in the BSW Com components.
- The COM module now supports AUTOSAR requirement [SWS_Com_00833].
- DoIP now allows DoIPTcpConnectionSecurityRequired parameter to be overridden at runtime, for cases where TLS certificate generation needs to be invoked via a diagnostic routine.
- The RTA-BSW Modules Xfrm and Rips can now be configured for 64 bit architectures using the new EcucParameters 'rba_Xfrm_64BIT_MachineSupport' and 'RteRipsCfgMachineSupport64Bit' respectively. In previous versions this behavior required overriding the C preprocessor macros 'RBA_XFRM_CFG_MACHINE_SUPPORT_64_BIT' and 'RTE_RIPS_CFG_MACHINE_SUPPORT_64_BIT'.
- The RTA-CAR Lin stack now supports auto addressing according to ISO 17987-3.

3.4.6 Version 12.3.0

- RTA-CAR on Linux now available, supporting Ubuntu 20.04 and 22.04.
- TLS 1.2 support.
- Addition of Chinese-market crypto primitives.
- Support for third-party crypto drivers.
- The rework of the Memory stack in line with AUTOSAR CONC_691.
- Security Extracts can now be configured and used with RTA-BSW Configuration Generation.
- Added support for Signal To Service Translation (using a non-AUTOSAR standard implementation) across the solution.
- Several extensions to RTA-SEC.
- CanTP now communicates with SoAd via PduR.
- Static ARP feature is supported in RTA-ETH.

3.4.7 Version 12.2.2

- Bugfixes in Dcm services OBD mode 9 and Read Data by Identifier (RDBI).
- LinIf change to address incorrect ECU response to invalid NADs.
- Fixed code generation time issues in CanTrcv for the TLE9255 tranceiver.

3.4.8 Version 12.2.1

- Fixed the install-time issue that triggered an unwanted computer restart or other unpredictable results during the RTA-CAR installation.

3.4.9 Version 12.2.0

- The RTA-A2L and MemMapGen code generators are now suitable for production use, these can be invoked from the RTA-CAR code generator UI on a compatible RTA-CAR project. Please refer to the RTA-CAR and respective tool documentation for additional details on the operation and features of these tools.
- Additional E2E profiles added to the RTA-Safe Stack, see the RTA-BSW documentation for details.
- The SOME/IP now includes an extension allowing for the transfer of data payloads in the Protobuf serialization format.
- Additional IdsM features, see the RTA-BSW documentation for details.
- Additional RTA-Sec stack features.

3.4.10 Version 12.1.0

- RTA-A2L and MemMapGen are included in RTA-CAR, these code generators can be invoked from the RTA-CAR code generator UI on a compatible RTA-CAR project. For this version these features should be used for evaluation purposes only as these generators are not suitable for use in production.

3.4.11 Version 12.0

3.4.11.1 RTA-CAR

- RTA-CAR supports Microsoft Windows® 11 host OS.
- RTA-CAR supports Microsoft Windows® Server 2022 host OS.
- Newly-created RTA-CAR projects are now based on the AUTOSAR R20-11 schema version.
- The RTA-CAR project creation wizard now includes extra steps when configuring an OEM project with GM-specific extensions.
- The RTA-CAR installation now provides both “RTA-CAR” and “ISOLAR-A/B Standalone” products. If you are upgrading from an earlier RTA-CAR or ISOLAR-A/B version, you may need to launch the product using a different shortcut or executable. See RTA-CAR User Guide, Section 2.1.6 ISOLAR-AB Standalone for further details.
- Additional RTA-CAR components, including OS ports, OEM extensions and custom ConfGen packages can now be installed directly from the RTA-CAR user interface. See RTA-CAR User Guide, Section 3.5 Updating RTA-CAR for further details.

3.4.11.2 ISOLAR-AB

- ComSignals and ComSignalGroups imported from a DBC file can optionally be suffixed with the name of the transmitting CAN bus.
- General bugfixes (see the ISOLAR-AB Release Notes for details).

3.4.11.3 RTA-BSW

- The KeyM module is now included as part of RTA-BSW.
- The RTA-CAR BSW modules have been upgraded to a new version, and now include a number of AUTOSAR R19-11 and R20-11 features. This is a major update with a large number of changes to configuration parameters from previous versions. If you require assistance with migration, please contact the RTA Hotline.
- ConfGen now supports the generation of a DCM EcucValue configuration from a DEXT file. For more details please see the RTA-BSW documentation.
- Improved ConfGen performance for some BSW modules.
- General bugfixes (see the RTA-BSW Release Notes for details).

3.4.11.4 RTA-RTE

- General bugfixes (see the RTA-RTE Release Notes for details).

3.4.12 Version 9.2.2

3.4.12.1 ISOLAR-AB

- General bugfixes (see the ISOLAR-AB Release Notes for details).

3.4.12.2 RTA-RTE

- The RTE uses a single IOC channel with multiple endpoints whenever a 1:n IOC communication is required. Use of 1:n IOC channel communication lowers the runtime and memory requirements in a multi-partition system.
- A bug that caused incorrect RTE code to be generated for COM-based transformation has been identified and fixed.
- General bugfixes (see the RTA-RTE Release Notes for details).

3.4.12.3 RTA-OS

- General bugfixes (see the RTA-OS Release Notes for details).

4 ETAS Contact Addresses

4.1 Technical Support

Technical support is available to all users with a valid support contract. If you do not have a valid support contract, please contact your regional sales office.

- https://www.etas.com/en/contact.php

The best way to get technical support is using the online support portal RTA Hotline:

- https://rtahotline.etas.com/

It is helpful if you can provide technical support with the following information (where relevant):

- Your support contract number
- The version of the ETAS tools you are using
- The version of the compiler tool chain you are using
- The command line (or reproduction of steps) that result in an error message
- The error messages or return codes you received (if any)
- Your RTA-CAR Project or if not possible the arxml file(s) containing the error(s).
- The *Diagnostic.dmp* if it was generated

4.2 ETAS HQ

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